Visual Intelligence & Integrations Research – NOVA Health

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Summary of recommendations. Read prescriptions with a multimodal LLM that returns structured data and make the user confirm it; treat food photos as estimates the user can edit; read watch data through HealthKit (iPhone, Apple Watch) and Health Connect (Android, Galaxy Watch) using the Flutter `health` package; and use Open-Meteo for weather, with WeatherAPI.com as backup.

- **OCR:** LLM vision extraction, with Tesseract and Google Vision as baselines. Published work reaches about 82% accuracy on drug names and doses, so human confirmation is essential.
- **Food:** dish recognition works well, but portion and hidden-ingredient estimates stay uncertain. The best published error is about 14% (needs depth images, Western meals), and a Peruvian-meal study found most estimates not accurate enough for consumer use.
- **Wearables:** Samsung's Data SDK is closed to new partners, so use Health Connect for Samsung; Apple needs a 99 USD per year developer account to publish. Galaxy Watch data arrives via the phone with a delay.
- **Weather:** Open-Meteo confirms all four variables and generous limits, but is non-commercial only.
- **Main risks:** handwriting errors, privacy of medical documents, and iPhone and Apple account access for testing.

Status: the wearable, weather and database facts were checked against official pages on 14 Oct 2026. The OCR test is a small first check on four synthetic documents (Tesseract and LLM only). Food photos were not tested; section 2 gives expectations from the literature.

1. OCR for prescriptions and reports

Proposed approach: send the photo to a multimodal LLM that returns structured JSON (medicine, dose, frequency, duration), then make the user confirm before anything is saved. Tesseract and Google Cloud Vision serve as comparison baselines. A small first test on four generated sample documents supports this: Tesseract failed on the blurred and handwriting-style samples, while the LLM read all of them.

What the literature shows

- Ali et al. (2024) combine Mask R-CNN (finds medicine regions) with TrOCR (reads the text), then match against a medicine database. They report a 1.4% character error rate, but their own test table shows 13.5%, so quote the test figure.
- Mankash et al. (2024, MIRAGE) fine-tuned three multimodal LLMs on about 743,000 simulated prescription images and reached about 82% accuracy on drug names and dosages. This supports the LLM-vision route, and also shows it is not close to perfect.
- Neither paper tests Tesseract or Google Vision on the same images, so the comparison below comes from our own small test.

Test table (synthetic sample documents; no real patients)

Image	Type	Tesseract	Google Vision	LLM (read by Claude)	Notes
1	Clear printed prescription, 2 medicines	8/8	Not run	8/8	Both read it perfectly
2	Printed, blurred with noise and tilt	0/8	Not run	8/8	Tesseract returned no text at all
3	Handwriting-style (computer simulation)	0/8	Not run	8/8	Tesseract output was garbled (for example "Me fori n D9 ti ce dqly"); the LLM read the dose "10 mg" with some doubt because the zero was distorted
4	Lab report with 4-column table, 3 tests	12/12	Not run	12/12	Both read the table correctly

How the test was done. Each cell is fields read correctly out of fields expected (medicine, dose, frequency and duration; for the lab report: test, result, unit and reference range). The four documents are synthetic samples generated for this test, with no real patients, and the handwriting sample is a computer simulation, not real handwriting. Tesseract was run for real with the supplied script (0.3 to 1.5 seconds per image, no internet needed). The LLM column is my own reading as a multimodal model; because I created the images it was not a blind test. Google Vision was not run because there was no access to it. Treat this as a small first check and repeat it with real sample prescriptions if your supervisor requires it.

Safety rule: the AI reads and suggests; it never changes a medicine or dose by itself. The user confirms every extracted field.

2. Food recognition and nutrition data

Recognising which dish is on the plate is mature; estimating how much is on the plate, and what is hidden in it, is still the weak point. NOVA should therefore show food and calorie values as estimates and let the user correct the portion.

Paper	What it shows	What it means for NOVA
Bianco et al. (2026), DINOv2 for food recognition	Frozen DINOv2 features match or beat state-of-the-art supervised methods on several food tasks, by up to +19.4% Top-1 accuracy on cross-dataset tests, while using a fraction of the parameters	One pretrained model can recognise dishes without training a model per task
Feng et al. (2026), RGB-D nutrient estimation	Best mean error (PMAE) of 14.43% on Nutrition5k (3,490 RGB-D images); mass error 9.91%, calorie error 13.80%; 0.44 s inference; needs depth images; fat has the highest error; global cuisines still need validation	Even the best result is about 14% off, and it needs a depth camera and was tested on Western cafeteria-style meals
Teng et al. (2027), survey of multimodal nutrition estimation	A single photo lacks metric scale (camera distance, container size, occlusion), and hidden ingredients and cooking methods are invisible; unconstrained deployment, portion estimation and cross-domain robustness remain open problems; datasets show cuisine imbalance	Supports labelling values as estimates and asking the user to confirm portion size
Dykstra et al. (2027), scoping review of 23 AI dietary-assessment studies	21 of 23 were image-based; main challenges were dataset variability, overlapping food items and light reflections	Real-world photos (shadows, mixed plates) will perform worse than benchmarks

Paper	What it shows	What it means for NOVA
Nakagawa & Yamamoto (2026), LLM nutrition estimation from food images	Tested three Claude models on Japanese meals, US meals and packaged foods; the stronger models clearly beat the smallest; packaged foods scored better than meals; salt estimates were consistently poor (MedAPE 34% to 64%)	Multimodal LLMs are a workable way to estimate nutrition from photos, but model choice matters and some nutrients are unreliable
Peruvian lunch-meal preprint (medRxiv, 2025), Gemma 3 models	On 510 Peruvian meals the best energy error was 108 kcal (27B model), with 45% of predictions within 100 kcal; predictions used text descriptions of the meals, not the photos; authors judged most estimates not accurate enough for clinical or consumer use	Closest evidence for non-Western meals: expect large errors on local dishes such as rice and curry
Antwi et al. (2026), meta-analysis of 262 AI/ML precision-nutrition studies	Modest pooled effects (body weight -1.97 kg, systolic blood pressure -4.05 mmHg); certainty of evidence low to very low; engagement and attrition matter	Do not overpromise health outcomes from food tracking alone

Nutrition data sources (checked on the official pages, 14 Oct 2026):

Source	Access	Limits	Licence
USDA FoodData Central	Free to use; data.gov API key needed (DEMO_KEY for trials)	1,000 requests per hour per IP; higher limits on request	CC0 public domain; credit requested
Open Food Facts	No key for reads; custom User-Agent required	Product reads 15 per minute, searches 10 per minute per IP (per user if requests come from a mobile app)	ODbL database; images CC BY-SA 3.0
Nutritionix / Edamam	Not checked	Not checked	Not checked

Local dishes such as rice and curry are likely thinly covered in these international databases, which matches the cuisine-imbalance point in Teng et al. and the Peruvian study. Plan manual entry or a small local food table as a fallback.

What to expect by photo type (from the literature; no food photos were tested for this version):

Photo type	Recognition	Portion and nutrient accuracy	Source
Packaged product	Expected good	Better than meal photos	Nakagawa & Yamamoto (2026)
Western-style mixed plate	Expected good	Best published error about 14%, with depth images	Feng et al. (2026)
Local rice and curry	Likely weaker	Large errors expected: best energy error 108 kcal on Peruvian meals, 45% of predictions within 100 kcal; salt estimates poor	Peruvian preprint (2025); Nakagawa & Yamamoto (2026)

Recommendation: identify the dish with a multimodal model, show the detected items for the user to confirm or edit, take the nutrition values from a food database, and display the result as estimated with an uncertainty note.

3. Wearable data: Apple HealthKit, Samsung and Health Connect

The proposal requires both Apple Watch and Samsung Watch data (steps, heart rate, sleep). Use HealthKit for iPhone and Apple Watch, and Health Connect for Android and Galaxy Watch, both

through the Flutter `health` package. Samsung's own Data SDK is not a practical route because it needs partner approval that Samsung is not currently granting. Checked on the official pages on 14 Oct 2026.

Option	Confirmed on the official page	Not confirmed
Apple HealthKit	Lets apps access and share health and fitness data with the user's privacy and control. Apple Developer Program costs 99 USD per year. A free Apple account can test on your own iPhone, but app IDs, devices and provisioning profiles expire after 7 days	The HealthKit page did not load for me, so check data types and storage yourself. I could not confirm whether HealthKit works with a free account
Android Health Connect	Android platform where apps store and share health data. Built in on Android 14 and higher; on Android 13 and lower the Health Connect app is installed from Google Play. The SDK supports Android 8 and higher, and the app needs Android 9 or higher. Data types include steps, sleep sessions and heart rate	SpO2 support (see the Data types index) and emulator testing (the page points to the Health Connect Toolbox)
Samsung Health through Health Connect	Samsung Health syncs with Health Connect (Samsung Health 6.22.5 and later). Apps can read steps and exercise, heart rate and sleep. Health Connect does not support Wear OS directly; Galaxy Watch data arrives through the phone's Samsung Health app on the watch's own sync schedule	Which Galaxy Watch models your team will use
Samsung Health Data SDK	Needs Samsung partner approval, and Samsung is not currently accepting Partner Apps Program applications. Needs a physical Android 6.0+ phone; no emulator support	Not a practical route for this project
Flutter `health` package (pub.dev)	Version 13.3.2, published 14 Aug 2026. Wraps HealthKit and Health Connect on Android and iOS. About 50 data types, many platform-specific. iOS 15 or later; Health Connect app must be installed on Android. Health Connect reads default to the last 30 days; on iOS the phone must be unlocked to read. Google Fit support was removed	Whether every data type your team needs works on both platforms

For each option record: data available (steps, heart rate, sleep), permissions required, cost, device requirements, and whether it can be demonstrated without owning the watch (for example with mock data).

Demo without owning a watch: use mock data in the app, and on Android use the Health Connect Toolbox that Google's documentation points to. HealthKit testing is easiest on a physical iPhone (I could not confirm this from Apple's page), so this depends on the supervisor's answer on iPhone access (section 6).

How published research uses wearable data: Merrill et al. (2025) describe PHIA, an LLM agent that answers questions about wearable data using code and multi-step reasoning. Khasentino et al. (2025, Nature Medicine) describe PH-LLM, a Gemini model fine-tuned on wearable sleep and fitness data. Both show that an AI engine can reason over wearable data once it is available, which is why this integration layer matters to the AGI Health Engine.

4. Weather API

Open-Meteo is the likely choice for the course prototype, with WeatherAPI.com as the backup. Open-Meteo's free API covers all four variables NOVA needs, but it is for non-commercial use only, so a commercial launch would need a paid plan. Checked on the official pages on 14 Oct 2026.

Provider	API key	Free limits	Temperature, humidity, rain probability, UV index	Catch
Open-Meteo	Pricing page does not say; test one request without a key	600 per minute, 5,000 per hour, 10,000 per day, 300,000 per month	Yes: hourly and daily forecasts list all four	Free use is non-commercial only
WeatherAPI.com	Yes, a `key` parameter on every request	100,000 calls per month; no card needed; 3-day forecast	Yes: forecast API documents temperature, humidity, chance of rain and UV (hourly and daily); no plan restrictions listed for these	Forecast length depends on plan
OpenWeatherMap	Sign-up and key	Free products: 60 calls per minute, 1,000,000 per month. One Call API: 1,000 calls per day free, then pay per call	One Call API returns temperature, humidity, UV index and rain probability (hourly and daily); pages do not say which fields the free products include	One Call needs its own subscription; card requirement not stated; pricing page labels One Call 3.0 deprecated and lists a newer 4.0, so check the current version

Decision: Open-Meteo first, because it confirms all four variables and generous limits for a prototype. Switch to WeatherAPI.com if you need a keyed service or a commercial plan later. Call the weather API from the backend, not from the mobile app, so keys and limits stay under your control.

Evidence that weather-based alerts are worth building: Heil et al. (2026, JMIR mHealth and uHealth) evaluated a heat early-warning app with 201 pregnant and postpartum women in South Africa over six months. Users rated usability highly and gained heat-health knowledge, but self-reported protective behaviour fell by the final assessment and attrition was substantial. For NOVA this means alerts should be personalised and not too frequent, to keep people engaged.

5. Feasibility risks

The biggest risks are wrong medicine data from handwriting, food values that look more precise than they are, and privacy of medical documents.

Risk	Impact	Mitigation	Evidence
Handwritten prescriptions read wrongly	Wrong medicine or dose saved	User confirms every field; match against a medicine list	Mankash et al.: about 82% accuracy even after fine-tuning
Portion size cannot be seen in one photo	Calorie and nutrient errors	Show values as estimates; let the user edit the portion	Teng et al.; Feng et al. (best error about 14%, with depth images)
Local dishes poorly covered	Wrong dish or nutrition values	Test with local meals; allow manual entry	Teng et al.; Feng et al. (Western meals only); Peruvian preprint (best energy error 108 kcal)
Some nutrients estimated badly	Misleading values such as salt	Hide or flag low-reliability nutrients	Nakagawa & Yamamoto (salt MedAPE 34% to 64%)
Real photos worse than benchmarks	Lower accuracy in practice	Guide the user to take clear photos	Dykstra et al. (lighting, overlapping foods)

Risk	Impact	Mitigation	Evidence
Samsung Data SDK needs partner approval	Cannot use Samsung's own SDK	Use Health Connect	Samsung developer page: applications not currently accepted
Galaxy Watch data arrives via the phone	Delayed or missing readings	Show last-synced time; do not promise live data	Samsung Health overview
Health Connect history limit	Only 30 days of past data by default	Request history permission or store data in the app database	pub.dev `health` package notes
No iPhone or paid Apple Developer account	HealthKit cannot be tested or published	Mock data; free account allows 7-day test builds on own device	Apple membership page (99 USD per year)
Weather free tier is non-commercial	Cannot launch commercially on free plan	Move to a paid plan or another provider before launch	Open-Meteo pricing page
Food database rate limits	Throttled or blocked requests	Cache results; call from backend	USDA 1,000 per hour; Open Food Facts 10 to 15 per minute per IP
Privacy of medical documents sent to LLMs	Legal and trust problems	Consent, encryption, send only what is needed, use fake documents in testing	Alzoubi & Mishra (2026): gaps in purpose limitation, data minimisation and lawful-basis documentation across LLMs
AI output trusted without checking	Unsafe decisions	Human confirms and AI only suggests	Chatzichristofis et al. (2026): keep the human as the decision-maker
Users stop engaging	Alerts and tracking lose value	Personalised, limited alerts	Heil et al. (attrition); Antwi et al. (low-certainty benefits)

6. Open questions for the supervisor

- Do we have a physical iPhone (and Apple Watch) for HealthKit testing, and is the 99 USD per year Apple Developer account available?
- Which Galaxy Watch and Android phone models will the team test with?
- Can we use real sample prescriptions, or only fake ones?
- What is the budget for LLM and OCR APIs?
- Should the food module target mainly local (Sri Lankan) meals, and is there a dataset for them?
- Since the Samsung Data SDK is closed to new partners, is Health Connect acceptable as the Samsung route?
- Where will medical documents be stored and who owns the privacy decisions?

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- OpenWeatherMap pricing (https://openweathermap.org/price) and One Call API (https://openweathermap.org/api/one-call-3)
- WeatherAPI.com: pricing (https://www.weatherapi.com/pricing.aspx) and docs (https://www.weatherapi.com/docs/)
- Android Health Connect (https://developer.android.com/health-and-fitness/guides/health-connect) and get started guide (https://developer.android.com/health-and-fitness/guides/health-connect/develop/get-started)
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- Samsung: Health Connect overview (https://developer.samsung.com/health/overview.html) and Health Data SDK process (https://developer.samsung.com/health/android/data/guide/process.html)
- Flutter health package (https://pub.dev/packages/health)
- USDA FoodData Central API guide (https://fdc.nal.usda.gov/api-guide) and Open Food Facts API (https://openfoodfacts.github.io/openfoodfacts-server/api/)

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